# View Reviews

**Paper ID**
258

**Paper Title**
Magnitude and Phase Aware Spectral Convolutional Neural Networks: Explainable Frequency-Domain Classification

**Track Name**
CVIP2026-Phase-I

**Reviewer #1**

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## Questions

**7. Please provide constructive and detailed feedback to help the authors improve their work. Comments should address strengths, weaknesses, and specific suggestions for improvement.**

The paper addresses an important challenge in convolutional neural network (CNN)-based image analysis by incorporating frequency-domain representations and explainability mechanisms. The proposed spectral learning framework demonstrates potential significance, particularly in medical imaging applications where interpretability and transparency are essential for clinical trust. The integration of FFT-based representations and activation-gradient fusion appears technically interesting. The main problem with this paper is that the literature review appears to have limited coverage of recent research publications. The absence of sufficient references to the latest studies makes it difficult to adequately compare the proposed work with current state-of-the-art approaches and assess its relative contribution to recent advancements in the field.

**Reviewer #2**

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## Questions

**7. Please provide constructive and detailed feedback to help the authors improve their work. Comments should address strengths, weaknesses, and specific suggestions for improvement.**

The paper presents a Magnitude and Phase Aware Spectral
Convolutional Neural Networks with Explainability. Overall, the contribution is useful.
However, some improvements are required as follows:

1. The contributions are not highlighted clearly in introduction. The motivation and novelty of the work is not clear.
2. The literature review needs more recent papers.
3. In the results, Table 1 and Table 2 are not clear. The proposed method is not mentioned and compared clearly.
4. Ablation study is needed.
5. Conclusion, future work needs improvement.

**Reviewer #5**

## Questions

**7. Please provide constructive and detailed feedback to help the authors improve their work. Comments should address strengths, weaknesses, and specific suggestions for improvement.**

The paper proposes a magnitude and phase-aware spectral CNN framework with explainability mechanisms, offering a potentially valuable contribution to medical image analysis. Kindly refer to the below review points.

1. The authors are encouraged to clearly highlight the motivation, novelty, and key contributions in the introduction of the paper. The literature review section also needs to be modified by positioning the identified research gaps against current state-of-the-art methods.

2. The proposed method simply looks a hybrid combination of spatial-to-frequency and then dual-frequency feature fusion. The actual contribution needs to be highlighted with a strong mathematical reasoning. What authors propose as a main contribution is missing in the proposed method. The same also needs to be reflected in the experiments section. A comparison with more methods and providing reasoning for the performance gaps can help to better understand the proposed method and contributions.